Technology Report and Presentation

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Student: Luis Guilherme de Barros Andrade Faria - A00187785

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Lecturer: Dr. Dyvia Leekha

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Table of Contents

1. Executive Summary 3
2. Benefits of Cloud Computing vs. Traditional IT Infrastructure 3
2.1. Cost Efficiency and Pay-as-you-grow Model 4
2.2. Rapid Scalability for Business Growth 5
2.3. Reduced IT Management Overhead 5
3. Challenges of Cloud Migration and Mitigation Strategies 6
4. Recommended Cloud Models for ABC Enterprise 7
5. Recommended Cost Model 8
6. Cloud Provider Recommendation: AWS 9
7. Conclusion 10
8. Appendices 12
8.1. Appendix A – Cloud vs Traditional IT Comparison Table 12
8.2. Appendix B – High-Level Architecture Diagram 12
8.3. Appendix C – Cost Model Breakdown Example 13
8.4. Appendix D – Shared Responsibility Model Diagram 13
8.5. Appendix E – NIST Five Essential Characteristics Mapped to ABC 13
8.6. Appendix F – Glossary of Terms 14
9. References 15

Cloud Computing for ABC Enterprise: Key Contributions to Business Automation

1. Executive Summary

Cloud computing has become a core business utility - email, storage, and smart devices all run on someone else's infrastructure, delivered on demand (Nishimura, 2022). For ABC Enterprise from our case study, this model translated directly into results: start-up IT costs dropped by ~80% and the platform absorbed a 10x customer surge within a month, without a matching increase in headcount (Eliaçık, 2022; McHaney, 2021). This report outlines three key benefits of cloud adoption, the main challenges to manage, and a practical recommendation for service model, deployment model, cost model, and cloud provider. Where ABC needs a provider beyond "XYZ", AWS is recommended for its maturity, global reach, and automation-ready managed services (Amazon Web Services, n.d.).

2. Benefits of Cloud Computing vs. Traditional IT Infrastructure

Traditional IT means owning everything - servers, cooling, procurement cycles, and the staff to keep it running (McHaney, 2021). For a high-growth start-up like ABC, that model is a strategic handicap. Cloud flips it: instead of buying capacity, ABC rents capability, aligned to the NIST essential characteristics of on-demand self-service, measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011).

2.1. Cost Efficiency and Pay-as-you-grow Model

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no idle hardware costs (Eliaçık, 2022). ABC's ~80% reduction in start-up IT costs is the "measured service" characteristic in action: pay for compute-hours, storage GB-months, and data transfer actually consumed (Mell & Grance, 2011). As McHaney (2021) notes, operational overhead - personnel, training, upgrades, and security - often exceeds hardware cost over time, making pay-as-you-go a structural advantage. Beyond the bill, standard tasks like backups, patching, and scaling can be codified and automated, reducing human toil across the delivery pipeline (Accenture Technology, 2020; Carpenter, 2020). Traditional IT forces a *lead strategy* - procuring capacity ahead of demand to avoid shortfalls - while cloud enables a *lag strategy* that provisions resources only as demand materialises, eliminating idle capital spend (McHaney, 2021). Figure 1 illustrates ABC's end-to-end request flow across the AWS stack.

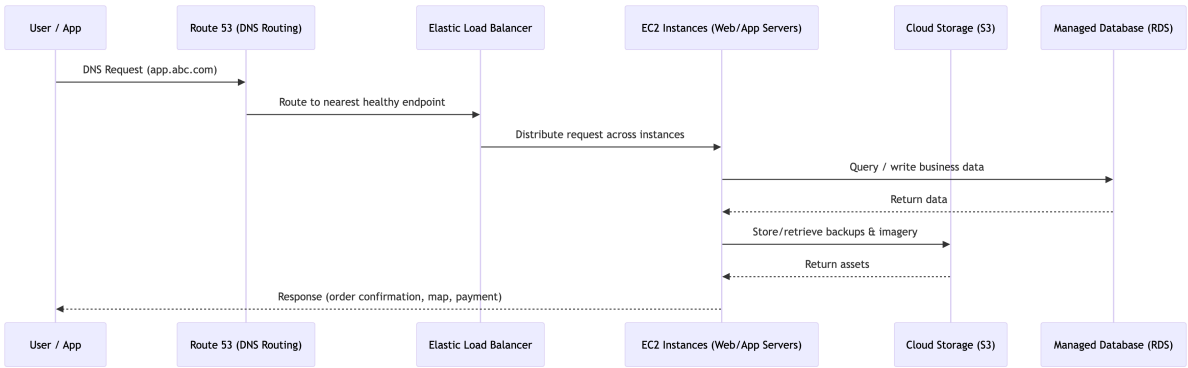

Figure 1 – ABC Enterprise cloud traffic flow - from user request through Route 53, load balancing, compute, storage, and database layers

2.2. Rapid Scalability for Business Growth

A 10x customer surge in a single month exposes the core weakness of on-premises infrastructure: procurement lead times mean hardware arrives after the opportunity has passed (McHaney, 2021). Cloud's rapid elasticity lets ABC scale compute up during campaigns or city launches and scale down when demand normalizes - automatically, not reactively (Mell & Grance, 2011). Eliaçık (2022) highlights that this "endless scalability" protects against outages from traffic spikes, keeping latency low and availability high. For an app handling delivery, taxi bookings, and payments, that's not a "*nice-to-have*" - it's a retention strategy. Cloud automation also strengthens resiliency - failed instances are automatically detected and replaced, removing single points of failure without operator intervention (Carpenter, 2020).

2.3. Reduced IT Management Overhead

In on-premises environments, more customers mean more infrastructure, which means more people to maintain it (McHaney, 2021). Cloud breaks that linear relationship. Through resource pooling and virtualization, providers consolidate physical resources across tenants - Manvi and Shyam (2021) describe this as partitioning, isolation, and consolidation that allows multiple workloads to share hardware without interference. ABC borrows that operational maturity: automated maintenance, managed monitoring, and resilient architectures that would be expensive to replicate in-house (Accenture Technology, 2020). Naved et al. (2022) confirm this pattern across diverse organisations: cloud resource pooling consistently reduces operational overhead while maintaining service quality.

3. Challenges of Cloud Migration and Mitigation Strategies

Cloud adoption is not risk-free. Three challenges are most relevant for ABC (Eliaçık, 2022):

1. **Security and Privacy:** ABC handles payments and customer PII. Security is the top concern for cloud adopters - 90% of security professionals cite it as a challenge, alongside governance (71%) and compliance (68%) (Manvi & Shyam, 2021). Eliaçık (2022) notes that entrusting sensitive data to a third party means breaches may not be detected immediately. Mitigation: least privilege IAM, mandatory MFA, encryption at rest/in transit, and continuous alerting. The shared responsibility model clarifies the division: the provider secures the cloud infrastructure - ABC secures what it deploys on top (Shore, 2020).
2. **Cost Volatility:** Pay-as-you-go can spiral without guardrails - overprovisioned instances and excessive egress generate surprise bills (Bittok, 2022). **Mitigation:** FinOps habits from day one: budget alerts, resource tagging, rightsizing, and reserved pricing for stable workloads.
3. **Vendor lock-in and Skills gap:** Deeper managed-service adoption makes provider migration expensive (Eliaçık, 2022), and cloud requires a different operating mindset - infrastructure-as-code, monitoring-first, shared responsibility (McHaney, 2021). **Mitigation:** prioritize portability (containers, standard databases) and invest in targeted upskilling. Manvi and Shyam (2021) highlight that chain dependency in virtualized environments is a key risk teams must understand to operate effectively.

4. Recommended Cloud Models for ABC Enterprise

Cloud service models sit on a control-versus-responsibility spectrum (IBM, n.d.-a; Linthicum, 2021; McHaney, 2021). IaaS gives ABC full flexibility over compute and configuration, but it also means managing operating systems and scaling policies. PaaS reduces that burden by abstracting infrastructure management, allowing the team to focus on building and improving the application itself. SaaS, while efficient, offers limited customization and is therefore less appropriate for a start-up that must differentiate its digital platform.

Deployment models also affect scalability and automation. NIST defines four models: public, private, hybrid, and community cloud - the last being shared among sector-specific organizations (e.g., banks, health systems) and not applicable to ABC's consumer platform (Manvi & Shyam, 2021). Private cloud increases control but sacrifices elasticity and cost efficiency. Hybrid cloud can support cloud bursting and workload portability, yet introduces architectural complexity that exceeds ABC's current operational maturity (Manvi & Shyam, 2021). For ABC - a fast-growing, customer-facing platform - public cloud remains the most strategically aligned option: Gartner analysis indicates IaaS workloads on public cloud experience 60% fewer security incidents than traditional data centers (IBM, n.d.-b), while delivering the scalability and built-in automation ABC requires (Mell & Grance, 2011; Nishimura, 2022) (see Table 1).

Deployment Model	Cost	Elasticity	ABC Fit
Public Cloud	Low (OPEX only)	High — auto-scales	✅ Recommended
Private Cloud	High (CAPEX + ops staff)	Limited — fixed capacity	❌ Over-engineered for start-up
Hybrid Cloud	Medium — dual infra	Moderate — complex to manage	⚠️ Premature — exceeds current maturity

Table 1 – Deployment model trade-offs for ABC Enterprise

Note on VPC: A Virtual Private Cloud, logically isolated private networking on public cloud infrastructure - offers a viable future option for ABC's more sensitive workloads, combining public cloud economics with enhanced network control (IBM, n.d.-c).

Recommendation: A blended approach: IaaS for flexible compute layers, and PaaS for managed databases, load balancing, autoscaling, and serverless functions. This balance preserves flexibility while embedding the rapid elasticity and measured service characteristics (Mell & Grance, 2011) directly into the infrastructure layer, without rebuilding operational maturity from scratch (Accenture Technology, 2020).

5. Recommended Cost Model

Cloud providers offer three levers:

1. Pay-as-you-go (maximum flexibility, highest unit price),
2. Reserved/committed pricing (discounts for baseline commitments),

3. Spot/preemptible (deep discounts for interruption-tolerant workloads) (Bittok, 2022; Amazon Web Services, n.d.-b).

Recommendation: A hybrid cost model - reserved capacity for stable customer-facing tiers (web/app, databases), pay-as-you-go autoscaling for demand spikes, and spot instances for background jobs and analytics pipelines. TCO modelling should include not just cloud fees but also migration cost, security tooling, and engineering effort (Bittok, 2022). As Bittok (2022) notes, cloud adoption is rarely about the cheapest bill - it's about better ROI: less downtime, faster launches, and automation that avoids linear headcount growth.

6. Cloud Provider Recommendation: AWS

Azure suits Microsoft-aligned enterprises; GCP leads in analytics. AWS best fits ABC: Route 53 already anchors the described architecture, and AWS offers the broadest automation-ready managed-service catalogue (Nishimura, 2022; Amazon Web Services, n.d.-a) (see Table 2).

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
AWS	Broadest managed-service catalogue	ELB — native Route 53 integration	Lambda — event-driven, zero idle cost	✅ Best fit — Route 53 already in stack
Azure	Microsoft / enterprise-aligned	Application Gateway — extra config needed	Azure Functions — strong but separate ecosystem	⚠️ No existing Azure signals in ABC
GCP	Analytics and ML-first	Cloud Load Balancing — GKE-oriented	Cloud Run / Functions — container-first	❌ No analytics-heavy workloads in ABC now

Table 2 – Cloud provider comparison for ABC Enterprise.

1) **Elastic Load Balancing:** Automates traffic routing and health-checks across EC2 instances without operator intervention - critical for ABC's delivery and taxi services where latency causes churn. ELB integrates natively with Route 53 and Auto Scaling already in ABC's stack; Azure Application Gateway and GCP Cloud Load Balancing require additional configuration to achieve the same (Amazon Web Services, n.d.).
2) **Auto Scaling:** Provisions or terminates EC2 instances based on demand signals, directly implementing NIST's rapid elasticity (Mell & Grance, 2011). The 10x surge required no manual intervention because capacity becomes policy-driven, not operator-driven - a fundamental shift from traditional IT (Amazon Web Services, n.d.).
3) **AWS Lambda:** Event-driven compute for ABC's workflow automation: order placed → delivery assigned; payment confirmed → restaurant notified. Unlike EC2, Lambda requires no server management; unlike SQS, no queue consumer. Functions scale to zero when idle and cost only per invocation (Amazon Web Services, n.d.; McHaney, 2021).

7. Conclusion

Cloud computing is the right strategic move for ABC: measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011) directly translate into business automation outcomes - scaling without procurement delays, operations without linear headcount growth, and faster time-to-market (Accenture Technology, 2020). The recommended stack is public cloud deployment, blended IaaS/PaaS service model, hybrid cost model, and AWS as provider. Managed well, ABC's infrastructure stops being a bottleneck and becomes a competitive advantage - scaling as

far as product-market fit allows (Eliaçık, 2022). That approach restructures operational capability into programmable, automated systems aligned with business growth.

8. Appendices

8.1. Appendix A – Cloud vs Traditional IT Comparison Table

Dimension	Traditional IT	Cloud (ABC Case)
Cost model	CAPEX heavy	OPEX elasticity
Scaling	Hardware procurement	Auto Scaling
Deployment speed	Weeks/months	Minutes
Automation	Manual ops	Policy-driven

8.2. Appendix B – High-Level Architecture Diagram

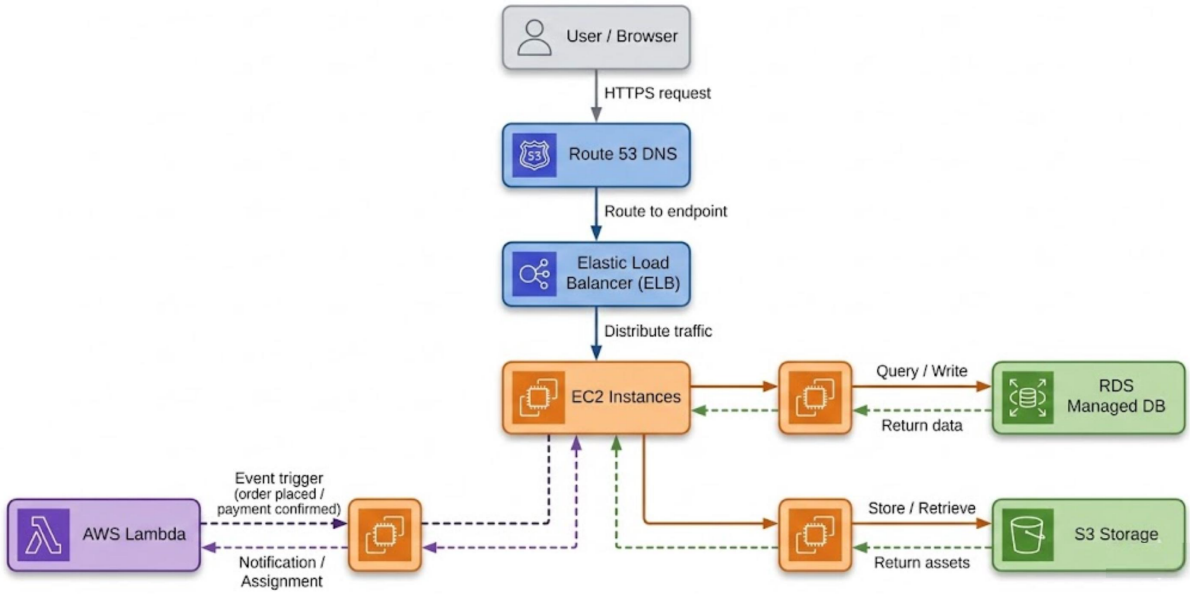

Figure 2 – ABC Enterprise high level cloud architecture on AWS.

8.3. Appendix C – Cost Model Breakdown Example

Cost Category	On-Premises	Cloud
Hardware	High upfront	None
Maintenance	Internal team	Managed
Scaling	Hardware purchase	Auto scaling
Downtime cost	High	Reduced

8.4. Appendix D – Shared Responsibility Model Diagram

Cloud Provider secures	ABC secures
Physical infra	IAM
Networking	Applications
Hypervisor	Data governance

8.5. Appendix E – NIST Five Essential Characteristics Mapped to ABC

NIST Essential Characteristic	ABC Enterprise Mapping
On-Demand Self-Service	Dev team provisions EC2/RDS via console — no vendor call needed
Broad Network Access	App accessible via mobile, browser, API across delivery + taxi regions
Resource Pooling	ABC shares AWS physical hardware; logically isolated via VPC
Rapid Elasticity	10× customer surge absorbed automatically via Auto Scaling
Measured Service	~80% cost reduction — pay only for compute-hours, GB-months consumed

Table 3 – NIST essential cloud characteristics mapped to ABC Enterprise (Mell & Grance, 2011).

8.6. Appendix F – Glossary of Terms

Term	Meaning
CAPEX	Capital Expenditure
OPEX	Operational Expenditure
IAM	Identity and Access Management
TCO	Total Cost of Ownership
SOA	Service-Oriented Architecture
PaaS	Platform as a Service
IaaS	Infrastructure as a Service
SaaS	Software as a Service

Table 4 – Glossary of Terms

End of Appendix Section

9. References

Accenture Technology. (2020, June 5). *Why cloud matters*. YouTube. https://www.youtube.com/watch?v=p1Nr03gtkyU

Amazon Web Services. (n.d.-a). *AWS Well-Architected Framework.* https://aws.amazon.com/architecture/well-architected/

Amazon Web Services. (n.d.-b). *AWS Pricing.* https://aws.amazon.com/pricing/

Bittok, T. (2022, January 22). *Cloud total cost of ownership.* LinkedIn Pulse. https://www.linkedin.com/pulse/cloud-total-cost-ownership-theophilus-bittok/

Carpenter, T. (2020, May 27). *AWS Certified Solutions Architect – Associate (SAA-C02) Cert Prep 1: Cloud services overview*. LinkedIn Learning. https://www.linkedin.com/learning/aws-certified-solutions-architect-associate-saa-c02-cert-prep-1-cloud-services-overview/benefits-of-cloud-computing

Eliaçık, E. (2022). *The good, bad, and ugly sides of cloud computing.* Dataconomy. https://dataconomy.com/2022/05/pros-and-cons-of-cloud-computing-2022/

IBM. (n.d.-a). *SaaS, PaaS, IaaS explained.* IBM Cloud. https://www.ibm.com/cloud/learn/iaas-paas-saas

IBM. (n.d.-b). *What is a public cloud?* IBM. https://www.ibm.com/think/topics/public-cloud

IBM. (n.d.-c). *What is private cloud?* IBM. https://www.ibm.com/think/topics/private-cloud

Manvi, S., & Shyam, G. K. (2021). *Cloud computing: Concepts and technologies* (Chapter 4). CRC Press. https://learning-oreilly-com.torrens.idm.oclc.org/library/view/cloud-computing/9781000338058/

McHaney, R. (2021). *Cloud technologies: An overview of cloud computing technologies for managers.* Wiley. https://ieeexplore-ieee-org.torrens.idm.oclc.org/servlet/opac?bknumber=9820907

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (Special Publication 800-145). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-145

Naved, M., Sanchez, D. T., Dela Cruz, A. P., Jr., Peconcillo, L. B., Jr., Peteros, E. D., & Tenerife, J. J. L. (2022). Identifying the role of cloud computing technology in management of educational institutions. *Materials Today: Proceedings, 51*(8), 2309–2312. https://doi.org/10.1016/j.matpr.2021.11.414

Nishimura, H. (2022, August 30). *Introduction to AWS for non-engineers: 1 cloud concepts*. LinkedIn Learning. https://www.linkedin.com/learning/introduction-to-aws-for-non-engineers-1-cloud-concepts-2/

Rahman, M. S., & Raza, M. (2021). Cloud computing security challenges and its potential solution. *International Journal of Computer Applications, 174*(6), 29–33. https://doi.org/10.5120/ijca2021921167

Shore, J. (2020). *Cybersecurity with cloud computing: Service models*. LinkedIn Learning. https://www.linkedin.com/learning/cybersecurity-with-cloud-computing-service-models

Singh, S., & Chana, I. (2016). Cloud resource provisioning: Survey, status and future research directions. *Knowledge and Information Systems, 49*(3), 1005–1069. https://doi.org/10.1007/s10115-016-0922-3